

Branch-wise versus clade-grouped gene-family evolution on the archaeal tree

Reconciliation of 7,059 gene families on the 257-taxon archaeal species tree, Euryarchaeota root

Reconciling each gene family against the species tree under a duplication–transfer–loss (DTL) model requires a choice of how finely rates may vary across the tree. We compared two parameterisations: the *clade-grouped* model, in which all branches of a major clade share a single set of duplication, loss and transfer rates (and origination weight), and a *fully branch-wise* model, in which every branch carries its own rates. As a control, our GPU reconciliation engine reproduces the reference per-family likelihoods to within Pearson $r = 0.999998$, so the comparison reflects the rate model, not the implementation.

Relaxing the clade constraint improves the fit by $\sim 10,900$ log-likelihood units (Table 1) while leaving the inferred root unchanged: Euryarchaeota remains the maximum-likelihood rooting under both models.

A model with thousands of free rates risks describing noise rather than signal, so we tested it by cross-validation (Figure 1). We estimated the rates from one half of the gene families and used them to predict the likelihood of the other half, which took no part in fitting. The branch-wise model predicts these withheld families better than the clade-grouped model, by $+0.44$ log-likelihood units per family — in the same direction, though smaller, as on the families used for fitting ($+2.71$ per family). Had the additional parameters been absorbing noise they would have *degraded* prediction on the withheld families; instead they improve it. The among-branch rate variation the model adds is therefore real, not an artefact of over-parameterisation.

That variation has a clear and interpretable structure (Figure 2). Duplication and transfer rates are essentially *clade-constant*: within each named clade the per-branch estimates scarcely depart from a single value (clade membership explains $R^2 \approx 0.94$ of their variance). Loss is markedly less clade-structured ($R^2 \approx 0.38$): its extra variation is distributed across branches rather than captured by clade identity. Biologically, gene gain (duplication, transfer) is well summarised by one rate per clade, whereas gene loss is more branch-idiosyncratic.

The reconstructed ancestral genome is robust to this added flexibility (Figure 3). Of the 7,059 families, the branch-wise model assigns 903 to the last archaeal common ancestor and the clade-grouped model 772; 722 are common to both (Jaccard 0.76), and the per-family probabilities of origination at the root agree closely ($r = 0.96$). The two models therefore recover essentially the same ancestral gene set, the branch-wise model reconstructing a modestly larger genome.

candidate root	clade-grouped log L	branch-wise log L	clade-grouped AU	branch-wise AU
Euryarchaeota	-1,642,425	-1,631,537	0.63	0.63
HalobacThermopl	-1,647,830	-1,635,105	<0.05	0.00
TackA	-1,643,485	-1,635,514	<0.05	0.00
MHH	-1,649,534	-1,637,704	<0.05	0.00
Kor	-1,647,973	-1,637,909	<0.05	0.00
TACA	-1,648,397	-1,638,428	<0.05	0.00
DPANN	-1,644,330	-1,639,707	<0.05	0.00
TAC	-1,648,893	-1,639,970	<0.05	0.00
Asgard	-1,649,022	-1,641,098	<0.05	0.00
MicraDia	-1,646,301	-1,642,346	<0.05	0.00
AMD	-1,647,588	-1,642,470	<0.05	0.00
UndineClu2	-1,647,958	-1,643,241	<0.05	0.00
Alti	-1,648,300	-1,643,347	<0.05	0.00
Cluster2	-1,648,352	-1,643,727	<0.05	0.00
Micra5	-1,648,861	-1,644,060	<0.05	0.00

Table 1: All 15 candidate roots under matched origination (both models include the origination term): clade-grouped versus fully branch-wise per-branch rates. Log-likelihoods are on gpurec’s marginal scale, directly comparable, and the approximately-unbiased (AU) supports are scale-invariant bootstrap values. Both models place Euryarchaeota first and reject every alternative root (AU < 0.05); the branch-wise model fits better at every root (by +10,888 at Euryarchaeota; held-out cross-validation +1,556, Figure 1). So the per-branch rates improve the fit throughout without changing the rooting. (AleRax’s parsimonious, BIC-best model omits origination and instead supports an {Euryarchaeota, MHH} region; it is the *origination* term — present in both models here — that resolves the rooting to Euryarchaeota, not the rate granularity. The two models also order the lower roots somewhat differently, reflecting that neither resolves them strongly.)

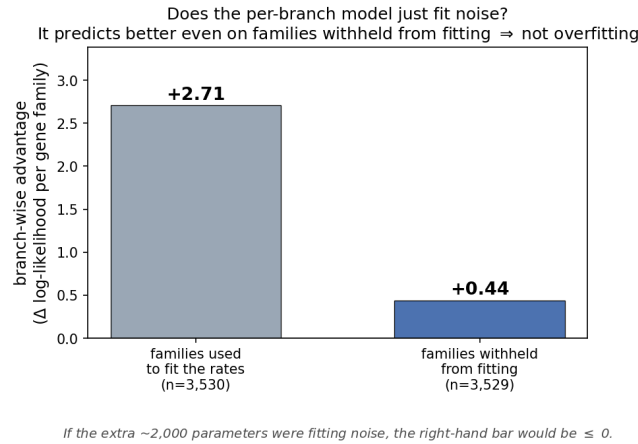


Figure 1: Cross-validation against over-parameterisation. The branch-wise model’s predictive advantage over the clade-grouped model, per gene family, on families used to fit the rates versus families withheld from fitting. Both are positive — the per-branch model predicts the withheld families better — so the added rates are real signal. Had they been fitting noise, the right-hand bar would be ≤ 0 .

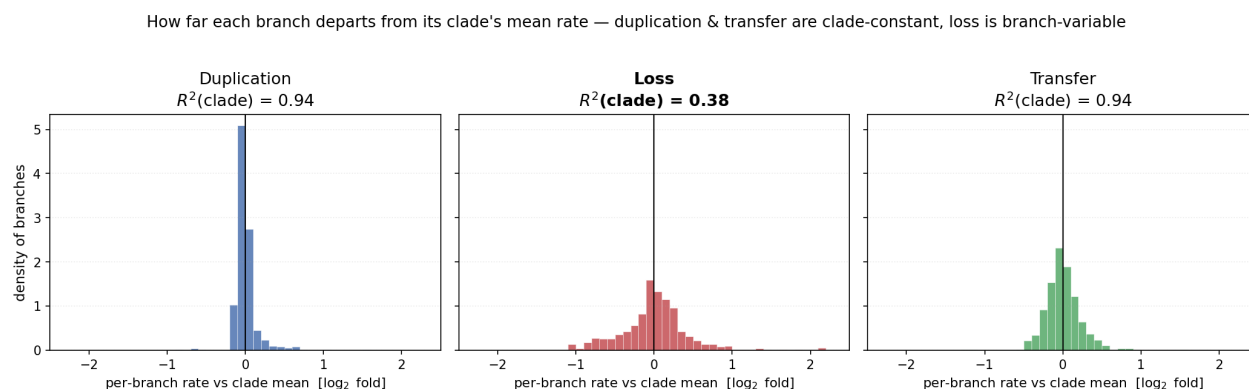


Figure 2: Departure of each branch's fitted rate from its clade's mean rate (log₂ fold; shared axes). Duplication and transfer form a sharp spike at zero — the per-branch estimates scarcely leave their clade mean, so clade membership explains $R^2 \approx 0.94$ of the variance — whereas loss is broad ($R^2 \approx 0.38$), i.e. branch-specific. Gene gain (duplication, transfer) is well summarised by one rate per clade; gene loss is not.

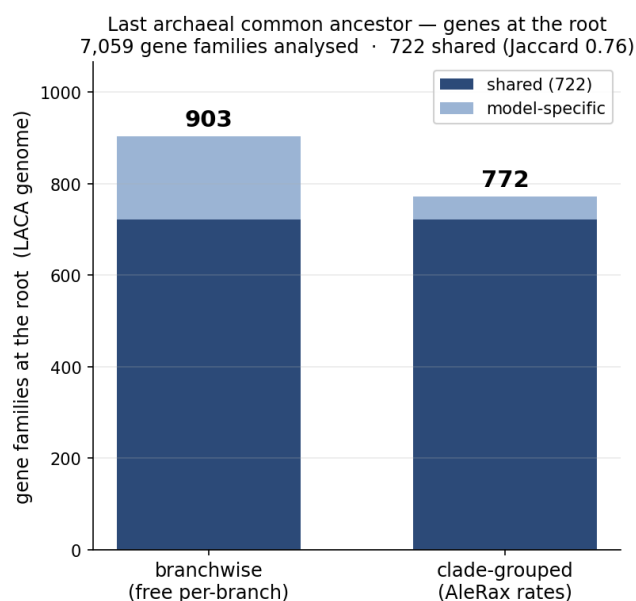


Figure 3: Reconstructed last archaeal common ancestor (LACA) gene content: 903 families under the branch-wise model vs. 772 under clade-grouped rates, 722 shared (Jaccard 0.76), of 7,059 analysed.